4.2 in Class Notes

Friday, October 17, 2025 9:34 AM

- Energy (9 forces)

Motion (Kinetic)

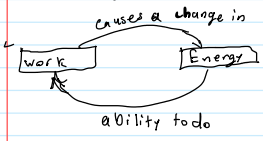
- Kinetic [E_k]
- Sound [E_{sound}]
- Radiant [E_{rad}]
- Thermal [E_{Th}]

Storage (potential) → Determined by location or configeration

- Gravitational potential [E_g]
- Chemical potential [E_{ch}]
- Electric potential [E_E]
- Nuclear potential [E_N]
- Elastic potential [E_e]

- As an object gets higher from the ground, the energy increases
- When you stretch out a rubber band, you change the configuration of the band's particles, which causes a change in energy

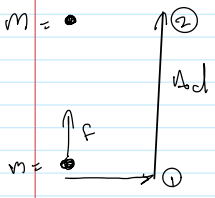
- $W = \Delta E$



?

- $E_k = \frac{1}{2}mv^2$
- $E_g = mgh$

If you dropped different objects from the same height, they would have different Energy because they have masses. The acceleration is gravity, which is different in someplaces.



$F = F_{net} = mg$

$W_f = F \cdot \Delta d$

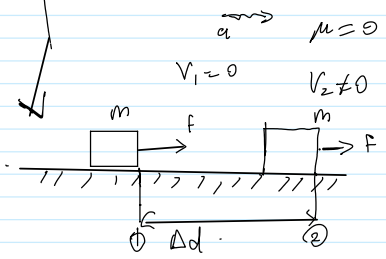
$(\theta = 0)$

$= F\Delta d$

$= mg\,\Delta d$

$\Delta E_g = mg\Delta d_2 - mg\Delta d_1$

$\Delta d = \Delta h$



$W_f = \vec{F} \cdot \vec{\Delta d}$

$W_f = ma\,\Delta d$

$V_2^2 = V_1^2 + 2a\,\Delta d$

$a\Delta d = \frac{1}{2}(V_2^2 - V_1^2)$

$W_f = m \times \frac{1}{2}(V_2^2 - V_1^2)$

$W_f = \frac{1}{2}mV_2^2 - \frac{1}{2}mV_1^2$

$= \Delta E_K$

- $W = F \cdot \Delta d \cos\theta$, $F_{net} = F$

$\theta = 0$

$W = ma\,\Delta d$.

- why is force or acceleration not a good a good way to measure energy?
  - ↳ You can have a force or acceleration without motion or displacement.

- change in energy = Energy 2 - Energy 1

- Conservation principal:
  - ↳ fundamental principal
  - ↳ first law of thermodynamis:
    - ↳ Law of the conservation of energy
    - ↳ Energy cannot be created or destroyed in an isolated system
    - ↳ It can only be transformed
    - ↳ Total energy in the universe is constant
    - ↳ When you drop a book the potential Energy turns to kinetic energy, which turns into thermal energy when it hits the ground (the ground' breifly heats up (slightly))

  - ↳ Systems (we only need 2 systems now because Einstein equated mass and energy)

    Open System

  - → Conservation of mechanical Energy
    - ↳ Mechanical energy (E_k, E_g) can't be created or destroyed in an isolated system that only has conservative forces doing work
    - ↳ mechanical energy can only be transformed to another type (mechanical to mechanical)

  - → Conservative forces vs Non - Conservative forces

Conservative forces	Non - Conservative forces
• Force is not path dependant	• Force is path dependant
• Force is always in 1 direction	• force is the direction of the moving object
↳ F_g: towards the center of another mass	• Ex → F_f = parallel to Surface
F_E: toward the center of another charge	
F_e: toward the equilibrium point	

$E_T = E_g + E_K$

Practice questions:

$E_K = \frac{1}{2} m v^2$

$v = 25\ m/s$

$m = 2\ kg$

$\mu = 0$

$E_K = \frac{1}{2} m v^2$

$E_K = \frac{1}{2}(2)(25)^2$

$= 625\ J$

$E = [J]$

$[kg][m/s]^2$

$1\ kg\ m^2/s^2 = 1\ J$

$W = [J]$

$W =$

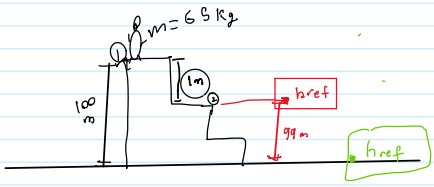
$[N][m]$

$[kg][m/s^2][m]$

$W = [kg][m/s]^2 = 1\ J$

$E_g = mgh$

always relative to a reference



different reference points

$E_{g_1} = mgh_1$

$= 65(9.8)(100)$

$= 63700\ J$

$E_{g_2} = mgh_1$

$= 65(9.8)(1)$

$= 637\ J$

↳ Def of Physics = Study of energy and matter